Robotic Physical-Property Recognition via Kinematic-Visual Fusion

Introduction and Methodology Draft

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I. Introduction

While computer vision has achieved remarkable success in the field of robotic perception, recent research has highlighted a critical deficiency known as "simplicity bias" in deep neural networks such as CNNs [1]. These networks tend to latch onto the most readily learnable features within the data, such as surface textures and chromatic patterns, and thereby neglect other physically significant characteristics. In robotic manipulation scenarios, this bias poses a substantial risk. Vision-dominated systems are prone to deception by objects that appear heavy and rigid but are physically soft and lightweight, or vice versa. This limitation underscores the nature of vision as a fundamentally "hypothetical" sense. To achieve genuine understanding, it is necessary to introduce a complementary "verifying" sense. This implies a modality capable of penetrating superficial appearances to provide authentic physical information, specifically through physical perception via object interaction.

Regarding physical perception, conventional robotic systems typically rely on tactile sensors or high-precision force/torque sensors [2]–[4]. Although effective, the high cost and fragility of these devices hinder their widespread adoption and application on low-cost platforms. An alternative approach utilizes proprioception, which leverages the internal signals of the robot actuators, including current, position, and velocity. However, interpreting these signals is non-trivial due to non-linear friction and high-frequency noise. Consequently, this method has historically necessitated complex analytical physical modeling [5].

This project proposes a paradigm shift from analytical modeling to data-driven methodologies. Our central hypothesis is that deep neural networks, particularly Transformer architectures, can directly extract implicit object properties like mass and stiffness from raw servo motor feedback without the need for explicit friction compensation. Furthermore, we posit that vision is a necessary but insufficient modality. While it provides essential semantic information, it fails when presented with deceptive objects. In contrast, proprioception is physically robust but lacks semantic context. Based on these premises, we propose a hybrid hard-sharing architecture to integrate these two modalities. By fusing visual features and proprioceptive signals at an early stage, this approach effectively resolves the problem of model arbitration during modality conflicts while simultaneously enhancing performance when multimodal information is congruent.

II. Methodology

In this section, we describe the methods used to achieve the project objectives.

A. Methodology Overview

The methodology of this project aims to rigorously isolate and evaluate the contribution of proprioceptive data in resolving visual ambiguities. Our experimental design circumvents the labor-intensive aspects of classical analytical robotics—specifically the reliance on precise kinematic calibration and explicit friction modeling—by relying entirely on the capacity of deep neural networks to extract latent information from raw signals.

To enable the neural network to learn the mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ between raw motor signals (Current, Load, Velocity, and Position) and physical properties, it is imperative to ensure that the signal variance σ^2_{ctrl} is attributable to the object properties rather than variations in the manipulator's motion. Therefore, we established a set of standardized manipulation primitives, comprising two primary actions: *Grasping* and *Lifting*[1].

B. Standardized Grasping

Rather than employing explicit force control, which tends to be unstable on low-cost servos, we utilize a position-based impedance strategy. Let w denote the gripper aperture (the Euclidean distance between the fingertips). For our revolute gripper mechanism, this aperture is a geometric function of the servo angle θ. To induce the standardized impedance response, we command a target aperture w_{target} that is strictly smaller than the object's actual physical width w_{object} (i.e., $w_{target} < w_{object}$), artificially inducing a position error $e(t)$. At discrete time t, the servo's underlying PID controller generates a PWM duty cycle signal $d(t)$ based on this error. During the quasi-static equilibrium phase of contact, the proportional term typically dominates, and the motor current $I(t)$ is positively

[1]Note that for the Feetech STS3215 servos employed in this study, "Load" is defined as the PWM duty cycle, ranging from -100% to +100%.

correlated with both the PWM duty cycle and the output torque $\tau(t)$. Consequently, the observed current signal can be modeled as a function of stiffness:

$$I(t) \propto \tau(t) \approx k_{obj}(w_{measured}(t) - w_{object}) \quad (1)$$

For rigid objects, the servo position stalls rapidly, maintaining a high error $e(t)$ and causing rapid current saturation. Conversely, for compliant objects, as deformation occurs, the rate of change in error is lower, resulting in a gradual rise in current. This indicates that stiffness information is not lost but is encoded in the dynamic trajectory of the current evolution (dI/dt).

C. Standardized Lifting

Following the grasp, the manipulator executes a vertical lifting motion. To ensure signal consistency, joint motion is governed by a trapezoidal velocity profile with a maximum velocity clamped at v_{limit}, ensuring a distinct constant-velocity phase during the lift. According to rigid body dynamics, the joint torque τ comprises inertial, Coriolis, gravitational, and frictional terms:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) \quad (2)$$

During the constant-velocity phase, where the joint velocity is limited to v_{limit} and acceleration $\ddot{q} \approx 0$, the inertial term vanishes. The dynamics equation simplifies to:

$$\tau_{cruise} \approx G_{loaded}(q) + F(v_{limit}) \quad (3)$$

Since the friction force $F(v_{limit})$ remains relatively stable at constant velocity, and the gravitational term $G(q)$ is linearly related to the object mass m_{obj} (where $G \propto m_{obj}g$), the incremental current ΔI relative to the no-load baseline during the steady-state cruise is primarily determined by the object's mass. Simultaneously, the current peak during the acceleration phase provides supplementary information regarding object inertia.

The mathematical formulations presented above provide a theoretical guarantee of observability for our "data-driven" approach. Although raw motor signals are subject to interference from non-linear friction and PID integral drift, the target physical properties (k_{obj}, m_{obj}) are not stochastically unobservable; rather, they are deterministically encoded within the error-driven motor current and load signal trajectories.

D. Data Acquisition

For each data acquisition episode, a global camera captures an RGB image of the object prior to any interaction. Throughout the interaction phase, we query four distinct signal types $\{P_t, I_t, L_t, V_t\}$ from the six joint servos at a frequency of 200 Hz. The specifications for these signals are detailed below.

Current Position (P_t) is measured in steps (1 step $\approx$ minimal resolution). In the absolute position control mode, the maximum value corresponds to the maximum effective angle.

Current Speed (V_t) represents the feedback of the current motor rotation speed, measured in steps per second.

Current Load (L_t) corresponds to the voltage duty cycle of the current control output driving the motor, measured in 0.1% units.

Current Current (I_t) denotes the real-time current consumption, measured in units of 6.5 mA (e.g., a value of 500 corresponds to 500×6.5 mA $= 3250$ mA).

E. Data Processing

Training deep neural networks on finite robotic datasets is prone to overfitting, particularly when models memorize static background features and absolute color values rather than intrinsic physical properties. To mitigate this issue and the aforementioned "simplicity bias," we implemented rigorous data augmentation strategies.

1) Vision: We randomly perturb the brightness, contrast, and saturation of the input image $x \in \mathbb{R}^{H \times W \times 3}$ via color jittering techniques. Additionally, geometric augmentations, including rotation, translation, and cropping, are employed to achieve spatial invariance. These operations are applied dynamically during batch generation, effectively expanding the diversity of the training distribution without requiring additional physical data collection.

2) Proprioception: We adopt a sliding window technique to process continuous proprioceptive data. We define the proprioceptive state of the robot at time step t as the vector $\mathbf{s}_t$:

$$\mathbf{s}_t = [j_t^{(1)}, j_t^{(2)}, \ldots, j_t^{(6)}] \in \mathbb{R}^{24} \quad (4)$$

Here, $j_t^{(i)} \in \mathbb{R}^4$ represents the four-dimensional feature vector (current, load, velocity, position) of the i-th joint of the six-degree-of-freedom manipulator.

To capture temporal dynamics, we construct sliding windows of length $L = 256$. For a given batch size B, the tensor input to the Transformer is formalized as $\mathcal{X}_{tactile} \in \mathbb{R}^{B \times L \times D_{total}}$, where $D_{total} = 6 \times 4 = 24$ channels. This high-dimensional representation enables the attention mechanism to simultaneously capture cross-joint correlations and temporal dependencies.

F. Architecture & Baselines

1) Baseline A: Vision-only (ResNet-18):

Input: Global image captured prior to interaction.

Hypothesis: This model is expected to achieve high accuracy on standard objects. However, it is predicted to fail with near 100% confidence during adversarial evaluation (e.g., classifying a tinfoil-wrapped sponge as metal), thereby demonstrating the existence of simplicity bias.

2) Baseline B: Tactile-only (Transformer Encoder):

Input: Raw sliding window sequences of proprioceptive data.

Hypothesis: This model is immune to visual deception and capable of correctly identifying physical properties (e.g., light vs. heavy). However, it lacks semantic capability and cannot distinguish between objects that share similar physical properties but differ in visual appearance.

3) Fusion Model: To resolve the conflicts inherent in the baseline models, we propose a Token-Level Early Fusion architecture. To align heterogeneous modalities into a unified latent space, we introduce separate linear projection layers ϕ_v and ϕ_t.

Visual Tokens: Feature maps extracted by ResNet are flattened and projected as the sequence $T_v = \phi_v(f_{vis}(x_{img})) \in \mathbb{R}^{N \times d}$.

Tactile Tokens: Proprioceptive windows are projected as the sequence $T_t = \phi_t(\mathcal{X}_{tac}) \in \mathbb{R}^{L \times d}$.

Subsequently, tokens are concatenated along the sequence dimension to form the joint input Z_0, which is directly fed into the shared Transformer Encoder:

$$Z_0 = [T_v \oplus T_t] + P_{pos} \tag{5}$$

We hypothesize that the self-attention mechanism within the shared Transformer can learn a dynamic "Sensory Gating" strategy. Specifically, when visual tokens and tactile tokens imply conflicting information, the attention weight matrix α_{ij} will automatically down-weight the saliency of visual tokens. This mechanism allows the model to prioritize proprioceptive evidence, effectively performing a "Physical Override" when cross-modal conflicts occur regarding physical property estimation.

4) Multi-Head Output Layer: To facilitate a hierarchical understanding of the object's physical properties, the fused latent feature vector Z_{out} is forwarded to three independent Task-Specific Multilayer Perceptrons (MLPs), corresponding to mass, stiffness, and material prediction, respectively. This Multi-Head Design enables the model to decouple the decision boundaries for different attributes while sharing the underlying feature representations.

Table I details the configuration of the three task-specific heads.

TABLE I
CONFIGURATION OF MULTI-HEAD OUTPUT LAYERS

Head	Dim	Class Definition / Purpose
Mass Est.	$N = 4$	{Very_Low, Low, Medium, High} *e.g., Empty vs. Solid*
Stiffness Est.	$N = 4$	{Very_Soft, Soft, Medium, Rigid} *e.g., Sponge vs. Metal*
Material Class.	$N = 5$	{Sponge, Wood...} *Check "Semantic Confusion"*

The final set of outputs is defined as $\mathcal{Y} = \{\hat{y}_m, \hat{y}_s, \hat{y}_{mat}\}$. Each output is transformed into a probability distribution via the Softmax function to facilitate the computation of the multi-task loss.

G. Adversarial Evaluation Strategy

To systematically evaluate the model's robustness against "visual deception," we constructed a dedicated evaluation dataset: Y3P-ECD v5.0 (Embodied Counter-Deception Dataset). Unlike random object collections, this dataset stratifies test subjects into three cognitively adversarial tiers based on Visuo-Tactile Consistency. Table II summarizes the definition and purpose of each tier.

TABLE II
Y3P-ECD V5.0 DATASET EVALUATION TIERS

Tier	Representative Object & Properties	Evaluation Purpose
Anchors (Prior)	**SandBox_Foil** Vision: Metallic Haptic: Heavy/Rigid	**Baseline Performance:** Validates basic capabilities and establishes visual priors (Silver = Heavy).
Traps (Spurious)	**Sponge_Red** Vision: Red (Bias: Heavy) Haptic: Light/Soft	**Simplicity Bias Test:** Checks if model overfits to color bias (Red $\neq$ Heavy). Tests OOD generalization.
Conflicts (Arbitration)	**Sponge_Foil** Vision: Metallic (Like Anchor) Haptic: Light/Soft	**Physical Override:** Ultimate test of fusion. Vision says "Heavy" (Hallucination), Tactile says "Light". Model must trust Tactile.

1) Anchors: Establishing the Prior: These objects are prevalent in the training set and are designed to establish strong visual priors. They teach the model that "silvery reflective textures typically correspond to high rigidity and mass." The test accuracy on Anchor objects serves as the system's Baseline Performance, validating the model's foundational capabilities under normal conditions.

2) Traps: Spurious Correlation Test: During the training phase, we deliberately introduced a statistical bias (e.g., a high prevalence of red objects being heavy and blue objects being light). Sponge_Red represents a quintessential OOD (Out-of-Distribution) sample; it retains the visual features found in the training set (red color) but inverts the physical properties.

This tests whether the system has overfitted to spurious correlations between color and mass (i.e., Simplicity Bias). If the model predicts the red sponge as "heavy," it indicates reliance on mere color matching. Conversely, a correct classification of "light" demonstrates the model's ability to utilize tactile signals to rectify statistical biases.

3) Conflicts: Cross-Modal Arbitration Test: This represents the most challenging adversarial sample. The object is visually indistinguishable from the anchor object (SandBox_Foil)—sharing identical texture features—but exhibits diametrically opposed behaviors during physical interaction. This creates a perfect Sensory Conflict.

This serves as the ultimate test of the Hard-Sharing architecture's arbitration capability. A successful prediction (identifying it as "deceptive/soft") implies that the attention mechanism within the Transformer has successfully assigned higher priority to tactile tokens, executing a "Physical Override" against the strong visual hallucination.

H. Implementation Details

1) Optimization: The network employs Cross-Entropy Loss for training. We define the total loss function $\mathcal{L}_{total}$ as the weighted sum of the losses from three independent tasks:

$$\begin{aligned}\mathcal{L}_{total} = \lambda_m \mathcal{L}_{CE}(\hat{y}_m, y_m) + \lambda_s \mathcal{L}_{CE}(\hat{y}_s, y_s) \\ + \lambda_{mat} \mathcal{L}_{CE}(\hat{y}_{mat}, y_{mat})\end{aligned} \tag{6}$$

Here, $\hat{y}$ represents the predicted logits (unnormalized probabilities), y denotes the ground truth labels, and λ represents the loss weights used to regulate the relative importance of different tasks. We utilize the AdamW optimizer with the weight decay rate set to 1×10^{-4} to prevent overfitting on the limited-scale robotic dataset.

2) Evaluation Metrics: We propose a novel metric to quantify the severity of simplicity bias, termed the **Deception Gap** (Δ_{Gap}). It is defined as the performance disparity between Anchor objects (normal) and Trap objects (deceptive):

$$\Delta_{Gap} = Acc_{Anchor} - Acc_{Trap} \tag{7}$$

A high Δ_{Gap} indicates an excessive model reliance on spurious visual correlations. This implies that the model identifies objects only when visual cues align with physical reality. Conversely, when $\Delta_{Gap} \approx 0$, it indicates robust arbitration capabilities and proves that model performance remains unaffected by visual deception.

Additionally, targeting the critical test case "Sponge_Foil," we perform a **Conflict Confusion Matrix Analysis**. We focus on monitoring the False Positive Rate (FPR) for the "Metal" category. A successful system must minimize off-diagonal errors. Specifically, this refers to instances where soft objects are erroneously classified as rigid bodies due to visual dominance.

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